The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Completion-Balance Ambient Dynamics
Necessity / Uniqueness from Zero-Hidden Selector Generator Law

Abstract

The current benchmark-facing theorem on the active primitive-reservoir observer-reduction line already sharpened the burden once more: the minimal zero-response affine completion core is no longer merely available, because it is derived canonically from a deeper completion-balance ambient dynamics package and is necessary there up to the obvious completion-state affine identifications. The remaining honest task is therefore no longer to justify the completion core itself. It is to justify why the completion-balance ambient dynamics package should hold.

The present note makes that next move in the strongest disciplined form now available without claiming final microphysical closure. For each sector it introduces a still deeper zero-hidden selector generator law consisting of a compact convex selector-state body, affine observer-data and shell-response readouts, a self-adjoint idempotent hidden selector, an exact selector-level body splitting over selector-zero and hidden directions, a positively gapped hidden-sector operator on the selector image, symmetry and time-reversal actions preserving the whole selector package, an exact zero-response shell realization on the selector-zero locus, fixed-data shell solvability on that same locus, and a coercive selector-law inequality on data-invisible directions modulo the inert zero-hidden kernel.

From that selector law one derives canonically every constituent of the completion-balance ambient dynamics package that now requires justification: the compact convex ambient admissible body, the affine observer-data and shell-response readouts, the invariant affine zero-hidden slice, the ambient balance operator with kernel equal to the zero-hidden translation space, the zero-hidden / hidden body splitting, the induced hidden factor and positive gap operator, the symmetry and time-reversal structure, the exact zero-response shell realization, fixed-data shell solvability, and the coercive balance inequality modulo the inert completion kernel. Within this selector-law class those constituents are not merely available. They are necessary and unique up to the obvious orthogonal identification preserving the selector splitting. The benchmark package remains fixed throughout: the one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has again narrowed what counts as an honest next step. The primitive-observer-reduction datum theorem, the defect-fiber factorization theorem, the one-metric output theorem, the chain-forcing theorem, the selector theorem, the stationary-construction theorem, the explicit-package theorem, the primitive-normal-form theorem, the ambient-shell-law theorem, the combined-response theorem, the graph-collapse theorem, and the later completion-core theorem each discharged what was honestly open at its own

layer. The latest result matters because it removes another tempting stopping point. Once the minimal zero-response affine completion core is itself a forced reduction of deeper completion-balance ambient dynamics, it is no longer enough to restate that ambient-balance layer under a different presentation.

The remaining honest burden is therefore one layer deeper again. The task is now to explain why the completion-balance ambient dynamics package should exist, rather than letting the ambient admissible body, the data and shell readouts, the zero-hidden slice, the balance operator, the hidden splitting, the shell realization, the solvability statement, and the coercive balance inequality stand as the new deepest disciplined assumptions. That is the burden addressed here.

The present note does not reopen the benchmark effective package. It does not alter the one operative completed metric, it does not introduce a second operative metric, and it does not enlarge the low-energy matter law. Its gain is narrower and more upstream. It introduces a still deeper zero-hidden selector generator law and proves that the completion-balance ambient dynamics package is the canonical and necessary reduction of that selector law on the active line.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not claim that final substrate microphysics has already been found. Its narrower theorem is only that, on the active primitive-observer-reduction line, the completion-balance ambient dynamics package is itself forced by a deeper zero-hidden selector generator law, and is unique there up to the obvious orthogonal identification preserving the selector splitting.

2. Fixed Primitive Continuation Data

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}},$$

a visible reduction map

$$\mathcal{R}_\sigma : \mathcal{I}_\sigma^{\mathrm{for}} \to X_\sigma,$$

and shell-level current and equilibrium carriers

$$\mathfrak{C}_\sigma^{\mathrm{cur}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{cur}}, \qquad \mathfrak{C}_\sigma^{\mathrm{eq}} : \mathcal{I}_\sigma^{\mathrm{for}} \to Z_\sigma^{\mathrm{eq}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}^{\mathrm{for}}_{\mathrm{car}} \times \mathcal{I}^{\mathrm{for}}_{\mathrm{cur}} \times \mathcal{I}^{\mathrm{for}}_{\mathrm{hom}}.$$

3. *Fixed shell extractors*

$$\mathscr{E}^{\mathrm{cur}}_{\sigma} : Z^{\mathrm{cur}}_{\sigma} \to Y^{\mathrm{cur}}_{\sigma}, \qquad \mathscr{E}^{\mathrm{eq}}_{\sigma} : Z^{\mathrm{eq}}_{\sigma} \to Y^{\mathrm{eq}}_{\sigma},$$

together with the recorded shell composites

$$\mathcal{J}^{\mathrm{prim}}_{\sigma} = \mathscr{E}^{\mathrm{cur}}_{\sigma} \circ \mathfrak{C}^{\mathrm{cur}}_{\sigma}, \qquad \Theta^{\mathrm{prim}}_{\sigma} = \mathscr{E}^{\mathrm{eq}}_{\sigma} \circ \mathfrak{C}^{\mathrm{eq}}_{\sigma}. \tag{1}$$

4. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^{\sharp}, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T^{\mathrm{matter}}_{\mu\nu}[g^{\sharp}, \psi] \tag{2}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$.

Remark 1. *Definition 1 is not reopened here. The primitive shell data and the benchmark one-metric / ordinary-matter outputs remain fixed continuation data. The work begins one layer deeper again: why the completion-balance ambient dynamics package that now carries the active line should itself be forced.*

3. Target Ambient Dynamics

Definition 2 (Completion-balance ambient dynamics with inert completion kernel)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *A* completion-balance ambient dynamics package *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean ambient state space*

$$\mathcal{Z}_{\sigma},$$

the observer-data space

$$\mathcal{Y}^{\mathrm{obs}}_{\sigma} \equiv \widehat{X}_{\sigma} \times Z^{\mathrm{cur}}_{\sigma} \times Z^{\mathrm{eq}}_{\sigma},$$

an ambient shell-response space

$$E^{\mathrm{amb}}_{\sigma},$$

and a compact convex ambient admissible body

$$\mathcal{B}^{\mathrm{bal}}_{\sigma} \subset \mathcal{Z}_{\sigma}$$

with nonempty interior.

2. *An affine observer-data readout*

$$\Lambda^{\mathrm{dat}}_{\sigma} : \mathcal{Z}_{\sigma} \to \mathcal{Y}^{\mathrm{obs}}_{\sigma}$$

and an affine shell-response readout

$$\Lambda_\sigma^{\mathrm{sh}} : \mathcal{Z}_\sigma \to E_\sigma^{\mathrm{amb}}.$$

3. *An invariant affine zero-hidden slice*

$$\mathcal{Z}_\sigma^0 \subset \mathcal{Z}_\sigma$$

with translation space

$$V_\sigma^0 \equiv T_z \mathcal{Z}_\sigma^0 \qquad (z \in \mathcal{Z}_\sigma^0),$$

and hidden complement

$$H_\sigma \equiv V_\sigma^{0\perp} \subset \mathcal{Z}_\sigma.$$

4. *A positive self-adjoint ambient balance operator*

$$\mathcal{K}_\sigma^{\mathrm{bal}} : \mathcal{Z}_\sigma \to \mathcal{Z}_\sigma$$

whose kernel is exactly the zero-hidden translation space,

$$\ker \mathcal{K}_\sigma^{\mathrm{bal}} = V_\sigma^0,$$

and whose restriction to the hidden complement obeys

$$\left\langle \eta, \mathcal{K}_\sigma^{\mathrm{bal}} \eta \right\rangle \geq \kappa_\sigma^{\mathrm{hid}} \|\eta\|^2 \qquad \textit{for all } \eta \in H_\sigma \tag{3}$$

for some $\kappa_\sigma^{\mathrm{hid}} > 0$.

5. *The hidden projection body*

$$D_\sigma \equiv \left\{ \eta \in H_\sigma : z + \eta \in \mathcal{B}_\sigma^{\mathrm{bal}} \textit{ for some } z \in \mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0 \right\}$$

is compact and convex, satisfies

$$0 \in \operatorname{int} D_\sigma,$$

and the ambient admissible body splits exactly as

$$\mathcal{B}_\sigma^{\mathrm{bal}} = \left\{ z + \eta : z \in \mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0, \quad \eta \in D_\sigma \right\}. \tag{4}$$

6. *A compact symmetry group*

$$\mathbb{B}_\sigma \subset O(\mathcal{Z}_\sigma) \times O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}})$$

and an orthogonal involution

$$\vartheta_\sigma^{\mathrm{bal}} \in O(\mathcal{Z}_\sigma) \times O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}})$$

preserving $\mathcal{B}_\sigma^{\mathrm{bal}}$, $\mathcal{Z}_\sigma^0$, $\Lambda_\sigma^{\mathrm{dat}}$, $\Lambda_\sigma^{\mathrm{sh}}$, *and* $\mathcal{K}_\sigma^{\mathrm{bal}}$.

7. *A smooth embedding*

$$\jmath_\sigma : X_\sigma \hookrightarrow \widehat{X}_\sigma$$

and a smooth zero-response shell realization

$$\zeta_\sigma^{\mathrm{bal}} : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{Z}_\sigma$$

whose image is exactly

$$\mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0 \cap (\Lambda_\sigma^{\mathrm{sh}})^{-1}(0),$$

and such that

$$\Lambda_\sigma^{\mathrm{dat}}(\zeta_\sigma^{\mathrm{bal}}(w)) = (\jmath_\sigma(\mathcal{R}_\sigma(w)), \mathfrak{C}_\sigma^{\mathrm{cur}}(w), \mathfrak{C}_\sigma^{\mathrm{eq}}(w)) \tag{5}$$

for every $w \in \mathcal{I}_\sigma^{\mathrm{for}}$.

8. *Fixed-data shell solvability on the zero-hidden slice:*

$$\Lambda_\sigma^{\mathrm{dat}}(\mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0) = \Lambda_\sigma^{\mathrm{dat}}\Big(\mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0 \cap (\Lambda_\sigma^{\mathrm{sh}})^{-1}(0)\Big). \tag{6}$$

9. *The inert completion kernel*

$$K_\sigma^0 \equiv \ker(D\Lambda_\sigma^{\mathrm{dat}}|_{V_\sigma^0}) \cap \ker(D\Lambda_\sigma^{\mathrm{sh}}|_{V_\sigma^0})$$

and the orthogonal completion-state projector

$$P_\sigma^{\mathrm{co}} : V_\sigma^0 \to K_\sigma^{0\perp} \cap V_\sigma^0.$$

10. *Completion-balance coercivity on data-invisible zero-hidden directions: there exists* $\kappa_\sigma^{\mathrm{co}} > 0$ *such that for every*

$$w \in \mathcal{I}_\sigma^{\mathrm{for}}, \qquad \delta z \in T_{\zeta_\sigma^{\mathrm{bal}}(w)}\mathcal{Z}_\sigma^0 = V_\sigma^0$$

satisfying

$$D\Lambda_\sigma^{\mathrm{dat}}\, \delta z = 0,$$

one has

$$\|D\Lambda_\sigma^{\mathrm{sh}}\, \delta z\|^2 \geq \kappa_\sigma^{\mathrm{co}}\, \|P_\sigma^{\mathrm{co}} \delta z\|^2. \tag{7}$$

Remark 2. *Definition 2 is the remaining layer singled out by the previous theorem. The present task is to derive or force the objects appearing there, not to restate the already-derived completion core.*

4. Zero-Hidden Selector Generator Law

Definition 3 (Zero-hidden selector generator law)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *A* zero-hidden selector generator law *for sector* σ *consists of the following data.*

1. *A finite-dimensional Euclidean selector-state space*

$$\mathcal{G}_\sigma^{\mathrm{sel}},$$

the observer-data space

$$\mathcal{Y}_\sigma^{\mathrm{obs}} \equiv \widehat{X}_\sigma \times Z_\sigma^{\mathrm{cur}} \times Z_\sigma^{\mathrm{eq}},$$

an ambient shell-response space

$$E_\sigma^{\mathrm{amb}},$$

and a compact convex selector-state body

$$\mathcal{B}_\sigma^{\mathrm{sel}} \subset \mathcal{G}_\sigma^{\mathrm{sel}}$$

with nonempty interior.

2. *An affine observer-data readout*

$$\Lambda_\sigma^{\mathrm{sel,dat}} : \mathcal{G}_\sigma^{\mathrm{sel}} \to \mathcal{Y}_\sigma^{\mathrm{obs}}$$

and an affine shell-response readout

$$\Lambda_\sigma^{\mathrm{sel,sh}} : \mathcal{G}_\sigma^{\mathrm{sel}} \to E_\sigma^{\mathrm{amb}}.$$

3. *A self-adjoint idempotent hidden selector*

$$\Sigma_\sigma^{\mathrm{hid}} : \mathcal{G}_\sigma^{\mathrm{sel}} \to \mathcal{G}_\sigma^{\mathrm{sel}}, \qquad {\Sigma_\sigma^{\mathrm{hid}}}^2 = \Sigma_\sigma^{\mathrm{hid}}, \qquad {\Sigma_\sigma^{\mathrm{hid}}}^* = \Sigma_\sigma^{\mathrm{hid}}.$$

Define the selector-zero locus

$$\mathcal{Z}_\sigma^0 \equiv \ker \Sigma_\sigma^{\mathrm{hid}},$$

its translation space

$$V_\sigma^0 \equiv \ker \Sigma_\sigma^{\mathrm{hid}},$$

and the hidden selector image

$$H_\sigma \equiv \operatorname{im} \Sigma_\sigma^{\mathrm{hid}}.$$

4. *A compact convex hidden body*

$$D_\sigma \subset H_\sigma$$

with

$$0 \in \operatorname{int} D_\sigma,$$

such that the selector-state body splits exactly as

$$\mathcal{B}_\sigma^{\mathrm{sel}} = \left\{ z + \eta : z \in \mathcal{B}_\sigma^{\mathrm{sel}} \cap \mathcal{Z}_\sigma^0, \quad \eta \in D_\sigma \right\}. \tag{8}$$

5. *A positive self-adjoint hidden-sector operator*

$$K_\sigma^{\mathrm{hid}} : H_\sigma \to H_\sigma$$

satisfying

$$\left\langle \eta, K_\sigma^{\mathrm{hid}} \eta \right\rangle \geq \kappa_\sigma^{\mathrm{hid}} \|\eta\|^2 \qquad \textit{for all } \eta \in H_\sigma \tag{9}$$

for some $\kappa_\sigma^{\mathrm{hid}} > 0$.

6. *A compact symmetry group*

$$\mathbb{S}_\sigma \subset O(\mathcal{G}_\sigma^{\mathrm{sel}}) \times O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}})$$

and an orthogonal involution

$$\vartheta_\sigma^{\mathrm{sel}} \in O(\mathcal{G}_\sigma^{\mathrm{sel}}) \times O(\mathcal{Y}_\sigma^{\mathrm{obs}}) \times O(E_\sigma^{\mathrm{amb}})$$

preserving $\mathcal{B}_\sigma^{\mathrm{sel}}$, $\Lambda_\sigma^{\mathrm{sel,dat}}$, $\Lambda_\sigma^{\mathrm{sel,sh}}$, $\Sigma_\sigma^{\mathrm{hid}}$, D_σ, *and* K_σ^{hid}.

7. *A smooth embedding*

$$\jmath_\sigma : X_\sigma \hookrightarrow \widehat{X}_\sigma$$

and a smooth zero-response shell realization

$$\zeta_\sigma^{\mathrm{sel}} : \mathcal{I}_\sigma^{\mathrm{for}} \hookrightarrow \mathcal{G}_\sigma^{\mathrm{sel}}$$

whose image is exactly

$$\mathcal{B}_\sigma^{\mathrm{sel}} \cap \mathcal{Z}_\sigma^0 \cap (\Lambda_\sigma^{\mathrm{sel,sh}})^{-1}(0),$$

and such that

$$\Lambda_\sigma^{\mathrm{sel,dat}}(\zeta_\sigma^{\mathrm{sel}}(w)) = (\jmath_\sigma(\mathcal{R}_\sigma(w)), \mathfrak{C}_\sigma^{\mathrm{cur}}(w), \mathfrak{C}_\sigma^{\mathrm{eq}}(w)) \tag{10}$$

for every $w \in \mathcal{I}_\sigma^{\mathrm{for}}$.

8. *Fixed-data shell solvability on the selector-zero locus:*

$$\Lambda_\sigma^{\mathrm{sel,dat}}(\mathcal{B}_\sigma^{\mathrm{sel}} \cap \mathcal{Z}_\sigma^0) = \Lambda_\sigma^{\mathrm{sel,dat}}\Big(\mathcal{B}_\sigma^{\mathrm{sel}} \cap \mathcal{Z}_\sigma^0 \cap (\Lambda_\sigma^{\mathrm{sel,sh}})^{-1}(0)\Big). \tag{11}$$

9. *The inert zero-hidden kernel*

$$K_\sigma^0 \equiv \ker(D\Lambda_\sigma^{\mathrm{sel,dat}}|_{V_\sigma^0}) \cap \ker(D\Lambda_\sigma^{\mathrm{sel,sh}}|_{V_\sigma^0})$$

and the orthogonal projector

$$P_\sigma^{\mathrm{co}} : V_\sigma^0 \to K_\sigma^{0\perp} \cap V_\sigma^0.$$

10. *Selector-law coercivity on data-invisible directions: there exists* $\kappa_\sigma^{\mathrm{sel}} > 0$ *such that for every*

$$w \in \mathcal{I}_\sigma^{\mathrm{for}}, \qquad \delta g \in T_{\zeta_\sigma^{\mathrm{sel}}(w)}\mathcal{G}_\sigma^{\mathrm{sel}}$$

satisfying

$$D\Lambda_\sigma^{\mathrm{sel,dat}}\, \delta g = 0,$$

one has

$$\|D\Lambda_\sigma^{\mathrm{sel,sh}}\, \delta g\|^2 + \left\langle \Sigma_\sigma^{\mathrm{hid}}\delta g, K_\sigma^{\mathrm{hid}}\Sigma_\sigma^{\mathrm{hid}}\delta g \right\rangle \geq \kappa_\sigma^{\mathrm{sel}} \left(\|P_\sigma^{\mathrm{co}}((I - \Sigma_\sigma^{\mathrm{hid}})\delta g)\|^2 + \|\Sigma_\sigma^{\mathrm{hid}}\delta g\|^2 \right). \tag{12}$$

Remark 3. *Definition 3 is smaller than the ambient package in exactly the way now needed. The zero-hidden slice and the ambient balance operator are not stipulated independently. They are generated by the selector* $\Sigma_\sigma^{\mathrm{hid}}$, *while the inert completion kernel is extracted from the selector-zero directions using the same data and shell readouts.*

5. Canonical Ambient Dynamics Induced by the Selector Law

Definition 4 (Canonical ambient dynamics induced by the selector law)**.** *Assume Definition 3. Define the canonical completion-balance ambient dynamics package by:*

1.

$$\mathcal{Z}_\sigma \equiv \mathcal{G}_\sigma^{\mathrm{sel}}, \qquad \mathcal{B}_\sigma^{\mathrm{bal}} \equiv \mathcal{B}_\sigma^{\mathrm{sel}}, \qquad \Lambda_\sigma^{\mathrm{dat}} \equiv \Lambda_\sigma^{\mathrm{sel,dat}}, \qquad \Lambda_\sigma^{\mathrm{sh}} \equiv \Lambda_\sigma^{\mathrm{sel,sh}}.$$

2.

$$\mathcal{Z}_\sigma^0 \equiv \ker \Sigma_\sigma^{\mathrm{hid}}, \qquad V_\sigma^0 \equiv \ker \Sigma_\sigma^{\mathrm{hid}}, \qquad H_\sigma \equiv \operatorname{im} \Sigma_\sigma^{\mathrm{hid}}.$$

3. The ambient balance operator is

$$\mathcal{K}_\sigma^{\mathrm{bal}}(v) \equiv K_\sigma^{\mathrm{hid}}(\Sigma_\sigma^{\mathrm{hid}} v), \qquad v \in \mathcal{Z}_\sigma. \tag{13}$$

4.

$$\mathbb{B}_\sigma \equiv \mathbb{S}_\sigma, \qquad \vartheta_\sigma^{\mathrm{bal}} \equiv \vartheta_\sigma^{\mathrm{sel}}, \qquad \zeta_\sigma^{\mathrm{bal}} \equiv \zeta_\sigma^{\mathrm{sel}}.$$

Proposition 1 (The selector law induces completion-balance ambient dynamics)**.** *Assume Definitions 3 and 4. Then the canonical data of Definition 4 satisfy every requirement of Definition 2.*

Proof. Because $\Sigma_\sigma^{\mathrm{hid}}$ is self-adjoint and idempotent,

$$\mathcal{G}_\sigma^{\mathrm{sel}} = \ker \Sigma_\sigma^{\mathrm{hid}} \oplus \operatorname{im} \Sigma_\sigma^{\mathrm{hid}} = V_\sigma^0 \oplus H_\sigma,$$

and the decomposition is orthogonal. Hence

$$H_\sigma = V_\sigma^{0\perp} \subset \mathcal{Z}_\sigma,$$

so the selector-zero locus is an invariant affine zero-hidden slice with translation space V_σ^0.

Equation (13) defines a self-adjoint positive semidefinite operator on $\mathcal{Z}_\sigma$. Its kernel is exactly $\ker \Sigma_\sigma^{\mathrm{hid}} = V_\sigma^0$, because K_σ^{hid} is positive definite on H_σ. For $\eta \in H_\sigma$,

$$\left\langle \eta, \mathcal{K}_\sigma^{\mathrm{bal}} \eta \right\rangle = \left\langle \eta, K_\sigma^{\mathrm{hid}} \eta \right\rangle \geq \kappa_\sigma^{\mathrm{hid}} \|\eta\|^2,$$

which is (3). The hidden projection body and the exact body splitting are exactly the hidden body D_σ and the selector-level splitting (8). Because D_σ is compact and convex with $0 \in \operatorname{int} D_\sigma$, the ambient splitting condition (4) holds.

The symmetry and time-reversal requirements follow immediately because $\mathbb{S}_\sigma$ and $\vartheta_\sigma^{\mathrm{sel}}$ preserve $\mathcal{B}_\sigma^{\mathrm{sel}}$, $\Lambda_\sigma^{\mathrm{sel,dat}}$, $\Lambda_\sigma^{\mathrm{sel,sh}}$, $\Sigma_\sigma^{\mathrm{hid}}$, and K_σ^{hid}, hence preserve the induced $\mathcal{B}_\sigma^{\mathrm{bal}}$, $\mathcal{Z}_\sigma^0$, $\Lambda_\sigma^{\mathrm{dat}}$, $\Lambda_\sigma^{\mathrm{sh}}$, and $\mathcal{K}_\sigma^{\mathrm{bal}}$.

The shell realization and data compatibility are exactly (10), because

$$\zeta_\sigma^{\mathrm{bal}}(\mathcal{I}_\sigma^{\mathrm{for}}) = \mathcal{B}_\sigma^{\mathrm{sel}} \cap \ker \Sigma_\sigma^{\mathrm{hid}} \cap (\Lambda_\sigma^{\mathrm{sel,sh}})^{-1}(0) = \mathcal{B}_\sigma^{\mathrm{bal}} \cap \mathcal{Z}_\sigma^0 \cap (\Lambda_\sigma^{\mathrm{sh}})^{-1}(0).$$

Fixed-data shell solvability (6) is exactly (11).

The inert completion kernel and projector are defined by the same formulas in the selector law and the ambient package, since $\Lambda_\sigma^{\mathrm{dat}} = \Lambda_\sigma^{\mathrm{sel,dat}}$, $\Lambda_\sigma^{\mathrm{sh}} = \Lambda_\sigma^{\mathrm{sel,sh}}$, and $V_\sigma^0 = \ker \Sigma_\sigma^{\mathrm{hid}}$. For the coercive

balance inequality, let

$$w \in \mathcal{I}_\sigma^{\mathrm{for}}, \qquad \delta z \in T_{\zeta_\sigma^{\mathrm{bal}}(w)} \mathcal{Z}_\sigma^0 = V_\sigma^0$$

with $D\Lambda_\sigma^{\mathrm{dat}}\, \delta z = 0$. Then $\Sigma_\sigma^{\mathrm{hid}} \delta z = 0$, so applying (12) gives

$$\|D\Lambda_\sigma^{\mathrm{sh}}\, \delta z\|^2 = \|D\Lambda_\sigma^{\mathrm{sel,sh}}\, \delta z\|^2 \geq \kappa_\sigma^{\mathrm{sel}} \|P_\sigma^{\mathrm{co}} \delta z\|^2.$$

Thus (7) holds with $\kappa_\sigma^{\mathrm{co}} = \kappa_\sigma^{\mathrm{sel}}$. All requirements of Definition 2 are satisfied. □

6. Forced Constituents and Uniqueness

Proposition 2 (All ambient constituents are forced by the selector law)**.** *Assume Definition 3. Then the following ambient-level constituents are fixed by the selector law itself.*

1. *The ambient admissible body is necessarily the selector-state body* $\mathcal{B}_\sigma^{\mathrm{bal}} = \mathcal{B}_\sigma^{\mathrm{sel}}$.
2. *The affine observer-data and shell-response readouts are necessarily* $\Lambda_\sigma^{\mathrm{dat}} = \Lambda_\sigma^{\mathrm{sel,dat}}$ *and* $\Lambda_\sigma^{\mathrm{sh}} = \Lambda_\sigma^{\mathrm{sel,sh}}$.
3. *The invariant zero-hidden slice is necessarily the selector-zero locus*

   $$\mathcal{Z}_\sigma^0 = \ker \Sigma_\sigma^{\mathrm{hid}},$$

   with translation space $V_\sigma^0 = \ker \Sigma_\sigma^{\mathrm{hid}}$.
4. *The hidden factor is necessarily the selector image*

   $$H_\sigma = \operatorname{im} \Sigma_\sigma^{\mathrm{hid}},$$

   the hidden body is necessarily D_σ*, and the positive gap operator is necessarily* K_σ^{hid}.
5. *The ambient balance operator is necessarily the selector-generated operator* (13)*; hence its kernel is forced to be the zero-hidden translation space.*
6. *The zero-hidden / hidden body splitting, the symmetry / time-reversal structure, the exact zero-response shell realization, fixed-data shell solvability, and the coercive balance inequality modulo the inert completion kernel are all induced by the same selector law and are not additional ambient freedom.*

Proof. Items (1) and (2) are immediate from Definition 4: once the selector-state body and the two readouts are fixed, no second ambient admissible body or second pair of ambient readouts is compatible with the same selector law. Item (3) follows because the only natural candidate for a zero-hidden locus produced by the selector law is the selector-zero locus, and the self-adjoint idempotent property forces its orthogonal complement to be the selector image.

Item (4) is exactly the selector image together with the hidden body and operator already present in Definition 3. Item (5) follows because the ambient balance operator must vanish exactly on selector-zero directions and reproduce the hidden-sector quadratic form on selector-image directions, which is precisely the action of $\mathcal{K}_\sigma^{\mathrm{bal}}(v) = K_\sigma^{\mathrm{hid}}(\Sigma_\sigma^{\mathrm{hid}} v)$. Item (6) is the content of Proposition 1: all those later ambient constituents descend directly from the same selector package and carry no additional independent freedom. □

Theorem 1 (Completion-balance ambient dynamics necessity / uniqueness from selector law)**.** *Assume Definition 3. Then:*

1. *the canonical construction of Definition 4 yields a completion-balance ambient dynamics package in the sense of Definition 2;*
2. *every ambient-level object singled out by the previous completion-core theorem is forced by the selector law: the compact convex ambient admissible body, the affine observer-data and shell-response readouts, the invariant affine zero-hidden slice, the ambient balance operator with kernel equal to the zero-hidden translation space, the zero-hidden / hidden body splitting, the induced hidden factor and positive gap operator, the symmetry and time-reversal structure, the exact zero-response shell realization, fixed-data shell solvability, and the coercive balance inequality modulo the inert completion kernel;*
3. *any other ambient package compatible with the same selector law differs from the canonical one only by the obvious orthogonal identification that preserves the selector splitting and the recorded data / shell coordinates;*
4. *consequently the completion-balance ambient dynamics layer is not an independent remaining freedom once the selector law is fixed.*

Proof. Item (1) is Proposition 1. Item (2) is Proposition 2. For item (3), any compatible reparameterization must preserve the orthogonal splitting

$$\mathcal{G}_\sigma^{\mathrm{sel}} = \ker \Sigma_\sigma^{\mathrm{hid}} \oplus \operatorname{im} \Sigma_\sigma^{\mathrm{hid}},$$

must preserve the selector law on each factor, and must leave the recorded data and shell coordinates unchanged. That leaves only the obvious orthogonal identification respecting the selector splitting. Item (4) is the routing consequence of the previous items. □

7. Benchmark Preservation

Theorem 2 (The derived ambient package preserves the same one-metric benchmark package)**.** *Assume Definition 3. Then the benchmark output statements of Definition 1 remain unchanged:*

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^\sharp, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T_{\mu\nu}^{\mathrm{matter}}[g^\sharp, \psi]$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$*. Consequently the deeper selector law and the induced ambient package introduce neither a second operative metric nor an enlarged low-energy matter law.*

Proof. The present note changes only the upstream origin of the ambient data. It does not alter the primitive shell $\mathcal{S}_{\mathrm{prim}}$, the recorded metric map $\mathscr{G}_{\mathrm{out}}$, or the recorded matter map $\mathscr{T}_{\mathrm{out}}$. Those remain the fixed continuation data of Definition 1, so (2) continues to hold exactly. □

8. Main Theorem

Theorem 3 (The remaining burden moves below the ambient-balance layer)**.** *Assume the fixed primitive continuation data of Definition 1 and, for each sector* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a zero-hidden*

selector generator law in the sense of Definition 3. Then:

1. *the completion-balance ambient dynamics package of Definition 2 is canonically induced by the selector law;*
2. *the compact convex ambient admissible body, the affine observer-data and shell-response readouts, the invariant affine zero-hidden slice, the ambient balance operator with kernel equal to the zero-hidden translation space, the zero-hidden / hidden body splitting, the induced hidden factor and positive gap operator, the symmetry and time-reversal structure, the exact zero-response shell realization, fixed-data shell solvability, and the coercive balance inequality modulo the inert completion kernel are all forced by that same selector law;*
3. *the one operative metric remains exactly $g^\sharp$, the ordinary matter route remains exactly the standard one through that metric, there is no second operative metric, and the low-energy matter law is not enlarged;*
4. *consequently the benchmark-facing burden after the previous completion-core theorem is discharged in the stronger form now available on the active line: the completion-balance ambient dynamics layer is no longer an unexplained stopping point, but a necessary reduction of a deeper zero-hidden selector generator law.*

Proof. Item (1) is Theorem 1(1). Item (2) is Theorem 1(2). Item (3) is Theorem 2. Item (4) is the combined routing consequence. □

Corollary 1 (What remains open after this theorem)**.** *After Theorem 3, the remaining honest burden is no longer to justify the completion-balance ambient dynamics package itself on the active primitive-observer-reduction line. The next honest burden is deeper again: derive or justify the zero-hidden selector generator law from still more primitive substrate dynamics, or else prove a sharper inevitability theorem that removes even that remaining selector-level freedom while preserving the same benchmark one-metric package and claim boundary.*

Proof. Theorem 3 converts the ambient-balance layer from a remaining benchmark-facing burden into a canonical reduction of a deeper selector law. What remains open is why that selector law itself should hold, rather than becoming the new disciplined stopping point. □

9. Conclusion

The positive result of this note is deliberately narrower than a completed final microphysical derivation and stronger than another deeper same-output relay. The previous theorem had already shown that the completion core is a forced reduction of completion-balance ambient dynamics. The present note goes one honest step further by showing that the ambient-balance layer itself is not left hanging either. It is canonically induced by a deeper zero-hidden selector generator law.

That gain directly addresses the precise objects that had become the next burden: the compact convex ambient admissible body, the affine observer-data and shell-response readouts, the invariant affine zero-hidden slice, the ambient balance operator with kernel equal to the zero-hidden translation space, the zero-hidden / hidden body splitting, the induced hidden factor and positive gap operator, the symmetry and time-reversal structure, the exact zero-response shell realization, fixed-data shell solvability, and the coercive balance inequality modulo the inert completion kernel. Within the

selector-law class those are no longer optional inserted ingredients. They are forced outputs, unique up to the harmless orthogonal identification preserving the selector splitting.

The benchmark boundary remains fixed throughout. The same primitive shell remains the shell carrying the observer outputs, so the one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. This is therefore a disciplined upstream derivational gain, not a final completion claim. The note still does not derive the selector law from ultimate substrate microphysics. That is the next honest open burden. But the completion-balance ambient dynamics layer is no longer the stopping point on the active line.